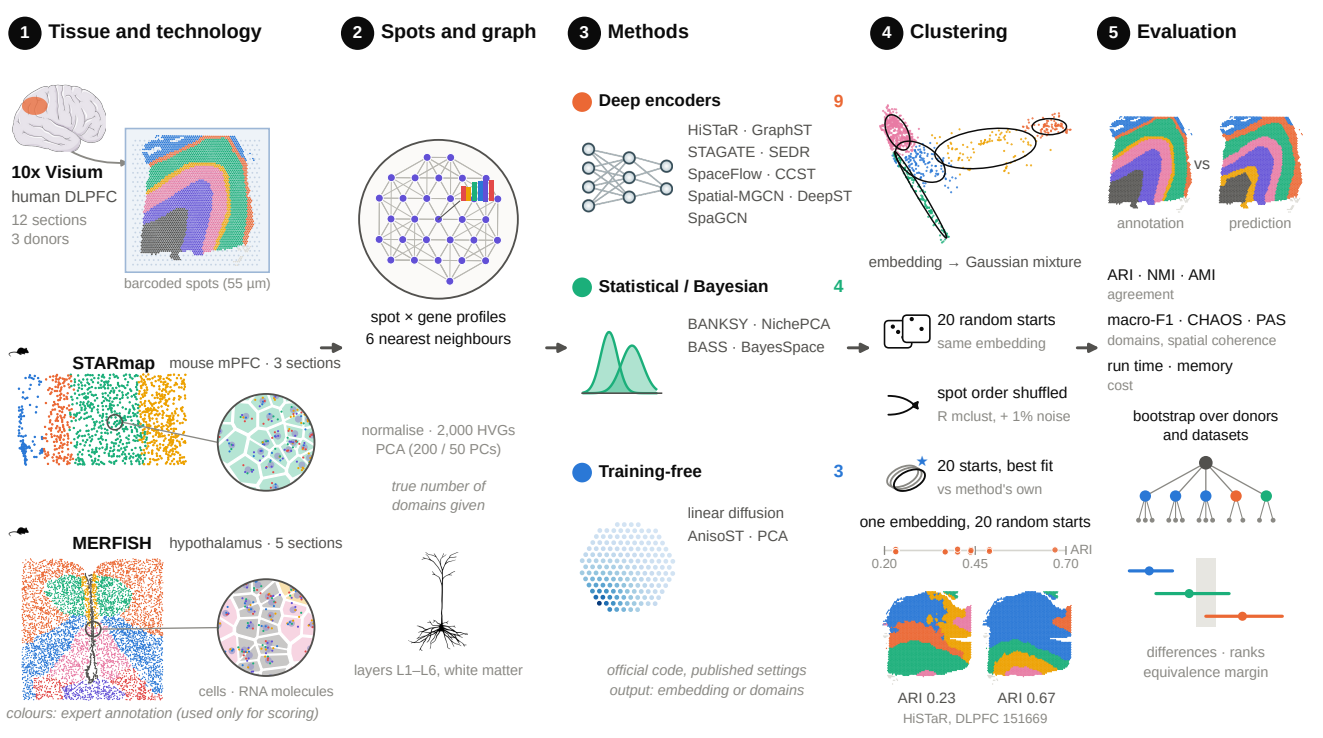

1 Tissue and technology
10x Visium
human DLPFC
12 sections
3 donors
barcoded spots (55 µm)
STARmap
mouse mPFC · 3 sections
MERFISH
hypothalamus · 5 sections
cells · RNA molecules
colours: expert annotation (used only for scoring)
2 Spots and graph
spot × gene profiles
6 nearest neighbours
normalise · 2,000 HVGs
PCA (200 / 50 PCs)
true number of domains given
layers L1–L6, white matter
3 Methods
Deep encoders 9
HiSTaR · GraphST
STAGATE · SEDR
SpaceFlow · CCST
Spatial-MGCN · DeepST
SpaGCN
Statistical / Bayesian 4
BANKSY · NichePCA
BASS · BayesSpace
Training-free 3
linear diffusion
AnisoST · PCA
official code, published settings
output: embedding or domains
4 Clustering
embedding → Gaussian mixture
20 random starts
same embedding
spot order shuffled
R mclust, + 1% noise
20 starts, best fit
vs method's own
one embedding, 20 random starts
ARI
0.20
0.45
0.70
ARI 0.23
ARI 0.67
HiSTaR, DLPFC 151669
5 Evaluation
vs
annotation
prediction
ARI · NMI · AMI
agreement
macro-F1 · CHAOS · PAS
domains, spatial coherence
run time · memory
cost
bootstrap over donors and datasets
differences · ranks
equivalence margin